PageWork

DD MM YYYY

(1)

(a) True

(b) ~~False~~ True

(c) T $y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(N-k)$

$y(n+N) = \sum_{k=-\infty}^{\infty} x(k)\, h(n+N-k) = y(n)$

(d) False poles of inverse & zeros of converse
but stable means poles in |z|<1

(e) True

(2) $x_c(t) = \sum_{k=-3}^{3} C_k e^{jk\omega_0 t}$

(a) $\omega_s \geq 2\Omega_{max}$ (Nyquist)

$> 6\omega_0$

(b) $\omega_n = \Omega T_s = \frac{k \cdot 4\pi}{9}$

$k=0 \Rightarrow \omega = 0$

$k=1 \Rightarrow \omega = \frac{4\pi}{9}$

$k=2 \Rightarrow \omega = \frac{8\pi}{9}$

$k=3 \Rightarrow \omega = \frac{4\pi}{3} - \frac{2\pi}{3}$

(3) $y(n) = \sum_{k=0}^{4} x(k)\, h((n-k) \bmod 5)$

$y(0) = x(0)h(0) + x(1)h(4) + x(2)h(3) + x(3)h(2) + x(4)h(1)$

($h(4)$ and $h(3)$ are 0)

$y(1) = x(0)h(1) + x(1)h(0) + x(2)h(4) + x(3)h(3) + x(4)h(2)$

($h(4)$ and $h(3)$ are 0)

$y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0)$

$y(3) = x(1)h(2) + x(2)h(1) + x(3)h(0)$

$y(4) = x(2)h(2) + x(3)h(1) + x(4)h(0)$

$y(2)$ & $y(3)$, $y(4)$ → same as linear conv

$y(1)$, $y(0)$ → aliased

(4) $G(e^{j\omega}) = H(-e^{j\omega}) \Rightarrow G(z) = H(-z)$

a) $-z_0$

b) $H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}$

$\Rightarrow H(-z) = \sum_{n=-\infty}^{\infty} h(n) (-z)^{-n}$

$= \sum_{n=-\infty}^{\infty} (-1)^n h(n) z^{-n}$

$G(n) = (-1)^n h(n)$

(5) $y[z] = -x[z] + z^{-1}x[z] - z^{-2}x[z]$

$= (-1 + z^{-1} - z^{-2}) x[z]$

a) $H(z) = -(1 - z^{-1} + z^{-2})$

$h(n) = -\delta[n] + \delta[n-1] - \delta[n-2]$

b) $H(e^{j\omega}) = -(1 - e^{-j\omega} + e^{-2j\omega}) = \frac{-(e^{2j\omega} - e^{j\omega} + 1)}{e^{2j\omega}}$

zeros: $e^{2j\omega} - e^{j\omega} + 1 = 0 \Rightarrow e^{j\omega} = \frac{1 \pm \sqrt{3}j}{2}$

$\Rightarrow \omega = \frac{\pi}{3}, -\frac{\pi}{3} \pm 2\pi k$

c) $\sum_{n=-\infty}^{\infty} |h(n)| = 3 \Rightarrow$ stable system

FIR

inverse poles lie on unit circle not inside

$\Rightarrow$ unstable

(6) $H(z) = \frac{k(z-4)(z-3)}{(z-0.5)} = \frac{k(1-4z^{-1})(1-3z^{-1})}{(1-0.5z^{-1})}$

$H_{min}(z)$ all poles & zeros inside $|z|=1$

$H_{ap}(z) \Rightarrow |H_{ap}(e^{j\omega})| = 1 \ \forall \omega \Rightarrow \left(\frac{z^{-1} - a^*}{1 - a z^{-1}}\right)$

$H_{min}(z) = \frac{(1-0.25z^{-1})(1-0.333z^{-1})}{(1-0.5z^{-1})}$

$H_{ap}(z) = \frac{z^{-1} - 0.25}{1 - 0.25z^{-1}} \cdot \frac{z^{-1} - 0.333}{1 - 0.333z^{-1}}$

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(8) a) poles @ 0.901 & 0.943 ⇒ close to 1
⇒ LPF
zero @ -0.95 close to -1

b) $H(z) = \frac{-44.07}{1-0.901z^{-1}} + \frac{45.07}{1-0.943z^{-1}}$

c) $H(z) = \frac{1+0.95z^{-1}}{(1-0.9z^{-1})-(1-0.95z^{-1})} = \frac{-44.05}{1-0.9z^{-1}} + \frac{45.05}{1-0.95z^{-1}}$

(6) $X[k] = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk}$

(a)

$X[0] = x(0)+x(1)+x(2)+x(3)$

$X[1] = x(0)+x(1)(-j)+x(2)(-1)+x(3)(j)$

$X[2] = x(0)+x(1)(-1)+x(2)(1)+x(3)(-1)$

$X[3] = x(0)+x(1)(j)+x(2)(-1)+x(3)(-j)$

$$F = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

(b) $x = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow y = Fx = \begin{bmatrix} 0 \\ -2j \\ 0 \\ 2j \end{bmatrix}$

(c) $z[n] = \frac{1}{16}\sum_{k=0}^{15} z[k] e^{j\cdot\frac{2\pi}{16}kn} = \frac{1}{4}\sin\left(\frac{\pi}{8}n\right)$